

Executive Summary

Enterprise Cloud Risk Management Assessment

Project Overview

This project applied the National Institute of Standards and Technology (NIST) Risk Management Framework (RMF) to a simulated government-hosted Amazon Web Services (AWS) environment supporting Human Resources (HR), Financial and Contract Information Network (FCIN), and Graduate Management Admission Testing (GMAT) applications.

The objective was to evaluate the system's security posture, identify risks, and develop governance and compliance documentation to support secure cloud operations.

Scope

The assessment focused on the AWS-hosted web infrastructure, including web servers, application servers, databases, identity and access management, audit logging, and continuous monitoring capabilities.

Key activities included:

- Security categorization
- Risk assessment
- Control assessment
- System Security Plan (SSP) development
- Audit logging strategy
- Continuous monitoring planning

Technologies and Frameworks

- NIST Risk Management Framework (RMF)
- NIST SP 800-53
- NIST SP 800-60
- AWS IAM
- AWS CloudTrail
- Amazon CloudWatch
- Amazon S3

Key Outcomes

The project identified security requirements for sensitive government information systems and developed governance-focused documentation supporting enterprise risk management objectives.

Major deliverables included:

- Security Categorization
- System Security Plan (SSP)
- AU-2 Audit Events implementation strategy
- Continuous Monitoring Strategy
- Risk assessment documentation

The proposed AWS architecture incorporated centralized logging, identity and access management, and monitoring capabilities designed to strengthen audit readiness and improve the organization's overall security posture.

Lessons Learned

This project strengthened practical knowledge of cloud governance, risk assessment, security documentation, and implementation of the NIST Risk Management Framework within a simulated enterprise environment. It also reinforced the importance of aligning technical controls with business and compliance objectives when designing secure cloud architectures.